

47. NEURAL CRESTS

DURATION : 11:21

STUDY SHEET

COURSE SUMMARY

This video delves into the crucial role of neural crests in embryological development, focusing on their impact on the formation of the peripheral nervous system, the face, and associated structures. Through key concepts such as cell migration and influence on the skin, it reveals how neural crests interact with the autonomic nervous system and affect organ health. Therapeutic implications are also explored, allowing therapists to reconcile anatomy, physiology, and epigenetics to better understand and treat their patients. The differentiation processes and epigenetic signals that guide neural crest formation are presented, offering avenues for innovative therapeutic approaches.

NEURAL CRESTS: IMPACT AND IMPLICATIONS

Neural crests are essential structures that appear during the closure of the neural tube. They extend along the neural groove and are crucial because they give rise to the entirety of the **peripheral nervous system**, including all nerves and ganglionic chains.

IMPACT OF NEURAL CRESTS ON THE ORGANISM

IN THE HEAD AND FACE

Intense **invasion** and **migration** of neural crests are responsible for the formation of a large part of the face, from bone to skin. At the base of the skull, tissue mixes and organises with the **mesoderm**, forming the **ectomesoblast**.

This migration is visible with the **rhombomeres**, which organise around the optic capsule and the mesencephalic area. Rhombomeres, numbered 1 to 7, invade the superior, median, and inferior pharyngeal arches.

THE SKIN

The **skin** is a tissue strongly influenced by the nervous system, partly derived from neural crests. It reacts to **solar radiation**, **pressure**, and **behaviour**. The face, in particular, reflects this state, allowing much information about internal health to be read.

THE AUTONOMIC AND VISCERAL NERVOUS SYSTEM

The ganglionic chain, the pre-visceral chain, and the intra-visceral ganglia originate from the neural crests. They also influence the **sympathetic** and **parasympathetic nervous systems**, informing organs such as:

- The heart
- The lungs
- The salivary glands (parotid, submandibular)
- The eye
- The digestive tract

Information from the neural crests can be visibly expressed on the face, allowing the internal state of organs such as the stomach or kidneys to be assessed. The quality of the skin, for example, reflects this information via the **dorsal ganglia** and the **enteric plexus**.

THE TRIGEMINAL NERVE AND CRANIAL NERVES

The migration of embryonic tissues, particularly neurological epithelial types, forms the ganglionic chain and participates in the formation of other structures through connections (integrins, laminins) promoting **melatonin**.

The **cranial nerves**, including the **trigeminal nerve**, whose 5 branches originate from this concentration, convey important information. A major branch transmits visceral information to the stomach and the vagus nerve.

Cranial distribution, including the **archeocranium**, **paleocranium**, **neurocranium**, and **neocranium**, is also conveyed by the neural crest. The formation of the trigeminal nerve depends on the **trigeminal ganglion** and a field that forms between the optic and nasal placodes. The trigeminal nerve results from three convergence fields induced by the dural sheaths, following information from the meninges.

OTHER IMPACTS OF NEURAL CRESTS

Neural crests influence the cascade development of various structures, such as the **inner ear**, **cartilages**, and **bones**. The eye, in its entirety, and its migratory face are independent of the neural crest. The latter positions itself at the level of the cornea, creating a neurological and facial continuity of mesodermal type, in relation to the entire body, including the heart.

DIFFERENTIATION AND EPIGENETICS PROCESS

It is essential to understand the processes of **differentiation**, **remodelling**, **proliferation**, **migration**, **delimitation**, **spiritualisation**, or **transduction**. This cascade of phenomena, although complex at the molecular level, has a direct impact on physiology.

Epigenetics plays a fundamental role in these processes, with specific signals that guide the development of the face and the formation of neural crests. These become **mesodermal** and then purely **neurological**.

They invade the **prosencephalon** (forebrain), **mesencephalon** (midbrain), and **rhombencephalon** (hindbrain), impacting the sympathetic and parasympathetic systems. All of this is linked to the neural crest.

ORIGIN AND THERAPEUTIC IMPLICATIONS

It is crucial to distinguish the embryological origins of different structures:

- **Base of the skull:** derives mainly from the **paraxial mesenchyme**.
- **Base of the dental arches:** derives mainly from the **neural crest**.
- **Spinal column:** derives from the ribs of the **paraxial mesenchyme**.
- **Limb skeleton:** derives from the **somatopleure** (limb buds).

These distinctions have major therapeutic implications:

1. **Face and teeth:** must be treated taking into account their origin from the neural crests.
2. **Base of the skull and spinal column:** their treatment must be homologous to their mesenchymal origin.
3. **Limb lesion:** involves the regulation of the peritoneum and the reintegration of the limb into the abdomino-thoracic cavity. This falls under a pulmonary plan and may require working on an apnoea brake at the lung level.

The articulation point between the **parietal peritoneum** and the **visceral peritoneum**, at the level of the pulmonary hilum, is an example of an area where neural crests play a role, acting as a suction field that invades the entire body.

This understanding of neural crest migration, particularly at the skull and shoulder level, offers avenues for therapeutic reflection.

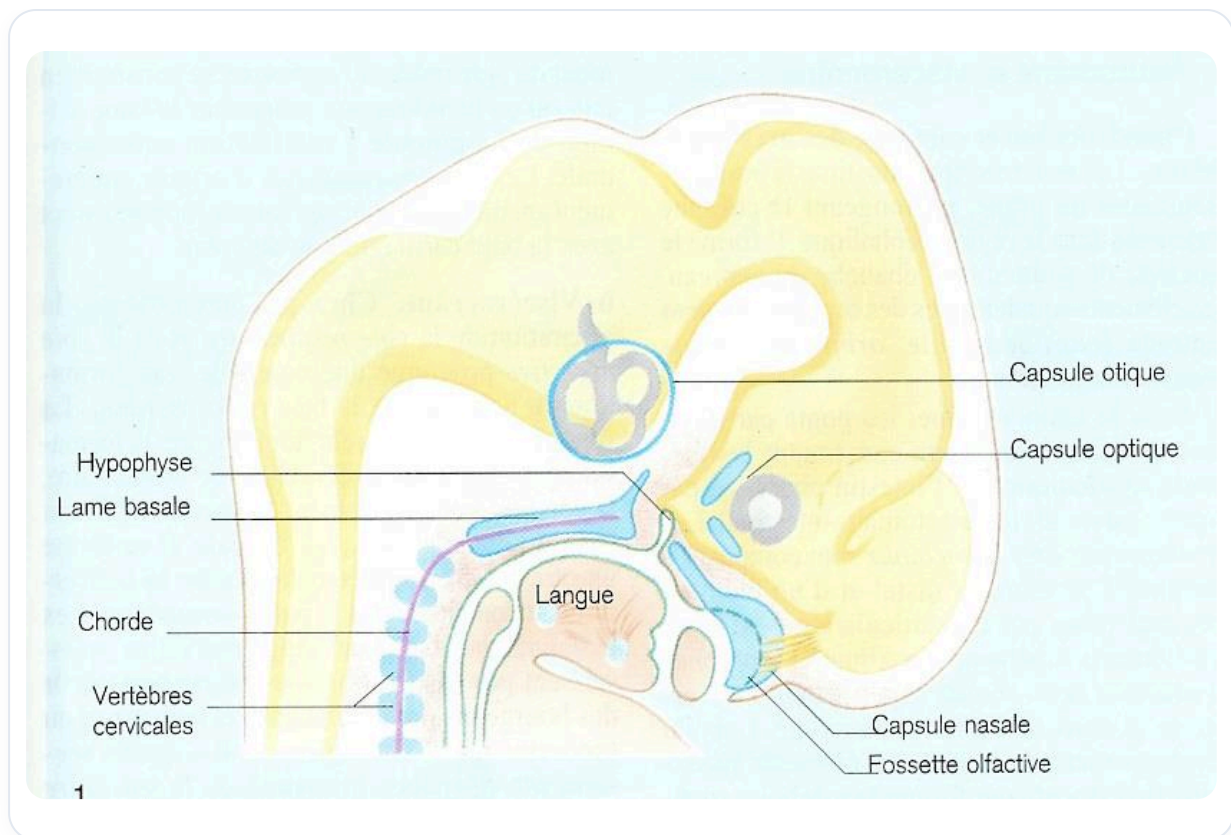


FIG. — *Human embryo sagittal*